

EXPERIMENT -2 (Merge Sort)

Name-Chitransh Phalkey

Prn:-25070521183(D)

CODE:-1.1.2

```
#include <stdio.h>
#include <stdlib.h>

void merge( int arr[], int left, int mid, int right)
{
    int n1 = mid - left + 1;
    int n2 = right - mid;
    int L[n1], R[n2];
    for (int i = 0; i < n1; i++)
        L[i] = arr[left + i];
    for (int j = 0; j < n2; j++)
        R[j] = arr[mid + 1 + j];
    int i = 0, j = 0, k = left;
    while (i < n1 && j < n2) {
        if (L[i] <= R[j]) {
            arr[k] = L[i];
            i++;
        } else {
            arr[k] = R[j];
            j++;
        }
        k++;
    }
    while (i < n1) {
        arr[k++] = L[i++];
    }
    while (j < n2) {
        arr[k++] = R[j++];
    }
}

void mergeSort(int arr[], int left, int right) {
    if (left < right) {
        int mid = left + (right - left) / 2;
        mergeSort(arr, left, mid);
        mergeSort(arr, mid + 1, right);
        merge(arr, left, mid, right);
    }
}

int main() {
    int n;
    scanf("%d", &n);
    int *arr = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    mergeSort(arr, 0, n - 1);
    for (int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    free(arr);
    return 0;
}
```

Screenshot:-Merge Sort(1.1.2)

CODETANTRA

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1.1.2. Merge Sort - Divide and Conquer

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You are given an array of n integers, and your task is to sort this array in non-decreasing order using the Merge Sort algorithm, which is based on the Divide and Conquer strategy.

Implement Merge Sort by defining the following user-defined functions:

- `mergeSort()` - recursively divides the array into subarrays until each subarray has a single element.
- `merge()` - merges two sorted subarrays into a single sorted array.

Input Format:

- The first line contains a single integer n representing the number of elements in the array.
- The second line contains n space-separated integers representing the elements of the array.

Output Format:

- Print the sorted array as a single line of n space-separated integers.

Constraints:

- $1 \leq n \leq 10^6$
- $-10^9 \leq \text{Array Elements} \leq 10^9$

Sample Test Cases

Explorer

C merge.c

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 void merge( int arr[], int left, int mid, int right) {
5
6     int n1 = mid - left + 1;
7     int n2 = right - mid;
8
9     int L[n1], R[n2];
10
11     for (int i = 0; i < n1; i++)
```

Average time

0.102 s

Maximum time

0.109 s

101.60 ms

109.00 ms

3 out of 3 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 99 ms

Expected output

5

9 -5 7 -2 -1

-5 -2 -1 7 9

Actual output

5

9 -5 7 -2 -1

-5 -2 -1 7 9

Test case 2 106 ms

Terminal

Test cases

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chitransh.phalkekeybatch2025@sitnagpur.siluedu.in

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